
Protein Pair-wise Sequence Alignment Webtool

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1. INTRODUCTION

This webpage provides a web-based tool for pair-wise protein sequence alignment service. Both a GLOBAL alignment using the Needleman-Wunsch algorithm and a LOCAL alignment using the Smith-Waterman algorithm are available. Unlike other ordinary alignments, users may select up to three substitution scoring matrixes with various weight for each matrix.

2. FEATURES

A. Alignment Algorithm

Two algorithms may be used. They employ substitution scoring matrixes instead of simple 'match/mismatch'.

- ① Needleman-Wunsch: A global alignment algorithm
- ② Smith-Waterman: A local alignment algorithm

B. Sequences

- ① Protein sequences may be pasted into the text box, or user can upload a text file containing a protein. Uploading a file has a higher priority than text area
- ② Non-amino acid characters will cause an error message. However, numeric characters and blank spaces are allowed and will be automatically removed.
- ③ A FASTA file or a sequence with a preamble line is also accepted.

C. Substitution Scoring Matrixes

- ① Users may select up to three different matrixes with respective weight.
- ② A new scoring matrix will be created and used depending on users' selection of matrix and weight
- ③ Currently standard BLOSUM and PAM matrixes are provided. Additional matrixes are hydrophobicity matrix and size matrix.
- ④ A weight value can be between 0 and 10. But at least one weight should be greater than zero. These weights are relative values, which mean that 1:1:1 is equal to 10:10:10.

D. Gap penalty

- ① Gap Open Penalty: To specify the opening gap penalty.
Default value is -10
- ② Gap Extension Penalty: To specify the gap extension penalty.
Default value is -1

E. Number of Maximum Alignments

- ① If more than one optimal alignment is found, user may set a limit to display to the higher ranked alignments. The default value is 3.

F. Result via Email

- ① Users may get the alignment result via email. Regardless of this option, users can see the results on the next display.

G. SAMPLE_SEQ Button

- ① This button will put two sample sequences into the sequence text boxes. They are variants of Drosophila Shaggy (Glycogen Synthase Kinase 3) proteins.

H. Result Download

- ① At the bottom of the result page, there will be three links for users to download the result files in text format. Please use right mouse button and save the target.

3. Clickable Buttons

- A. **ALIGN** : Begin to align two protein sequences
- B. **CLEAR** : Erase the user's input
- C. **SAMPLE_SEQ** : Put two sample protein sequences into the text area

4. Links

A. Main Page

- ① README : This read-me file
- ② Sample_Result : Sample alignment results with two shaggy proteins
- ③ L519 Bioinformatics : Simple introduction page to the class
- ④ Email : Send an email to the author

B. Result Page

- ① GLOBAL : Download the global alignment result
- ② LOCAL : Download the local alignment result
- ③ BOTH : Download both the global and local alignment results

5. Main page and sample result page

http://biokdd.informatics.indiana.edu - Protein Sequences Alignment Tool - Microsoft Internet Explorer

Protein Sequence Alignment Webtool

Hello! Welcome to Junguk's Sequence Alignment Webtool. Here you can align two protein sequences using either the Needleman-wunsch global alignment algorithm or the Smith-waterman local alignment algorithm. You may choose up to three scoring matrixes and their **RELATIVE** weights can be adjusted. Make sure that at least one of the three matrix weight should **NOT BE ZERO**.

ALIGNMENT METHOD	☒ Global Alignment (Needman-wunsch) ☒ Local Alignment (Smith-waterman)					
SEQUENCES Paste sequences or upload files. (Files have a higher priority)	Sequence#1 SORPFTTCSRDGSKITTVVATPGQGTDRVQEVSYTDTKVI FQAKLCDTGLVLAIKKVLQDRRFKNRELQIMRKLEHCNIVKLLYFFYS SOKRDEVFLNLVLEIPIETVYKVARFIRLYMYQLFRSLAYIHSLGIC HRDIKPNLLDPETAVLKLCDFGSAKOLLHGEFNVSYICSRYYRAPE LIPOADNYTTKIDVUSAGCVLAELLLOQIFFPDSDVDQLVEVIVLGO or Browse... Sequence#2 MSGPPTSSFAEGNKQSPSLVLGGVKTCSRDSKITTUVATPGQGTDR VQEVSYTDTKVI FQAKLCDTGLVLAIKKVLQDRRFKNRELQIMRKLEHCNIVKLLYFFYS SOKRDEVFLNLVLEIPIETVYKVARFIRLYMYQLFRSLAYIHSLGIC HRDIKPNLLDPETAVLKLCDFGSAKOLLHGEFNVSYICSRYYRAPE LIPOADNYTTKIDVUSAGCVLAELLLOQIFFPDSDVDQLVEVIVLGO or Browse...					
SCORING MATRIX	Matrix#1 BLOSUM62	Weight 10	Matrix#2 HYDROPHOBICITY	Weight 0	Matrix#3 SIZE	Weight 0
GAP PENALTY	Gap Opening Penalty : -10		Gap Extension Penalty : -1			
NUM OF MAXIMUM ALIGNMENTS	Maximum number of alignments to display, if more than one : 3					
Result to email	☐ Send the result to the following email address :					

Detailed Description : [README](#) [Sample Result](#)
Last Modified : Nov. 4, 2004
Class : [L519 Bioinformatics : Theory and Application](#)
Got any comment? Send an email to me

Protein Sequence Alignment Webtool

GLOBAL Alignment In Progress

```
#Alignment (Global) Successfully Completed
#Combined Scoring Matrix Created Using The Following
# Matrix1 : BLOSUM62 Weight: 100.0%
# Matrix2 : HYDROPHOBICITY Weight: 0.0%
# Matrix3 : SIZE Weight: 0.0%
#Seq1 Length: 482
#Seq2 Length: 575
#GapOpening Penalty: -10
#GapExtension Penalty: -1
#Number of Alignments Found: 2
#Number of Alignments Displayed: 2

>>Rank.1 Global Alignment Score:2349.0 Sum:731564.0 Identity:99.6%
seq1: 1 -SGRPRT-----TCSRDSKITTUVATPGQGTDRVQEVSYDTKVI
      |||||
seq2: 1 MSGRPRTSSFAEGNKQSPSLVLGGVKTCSRDSKITTUVATPGQGTDRVQEVSYDTKVI

seq1: 41 GNGSFGVVVFQAKLCDTGLVLAIKKVLQDRRFKNRELQIMRKLEHCNIVKLLYFFYSGEK
      |||||
seq2: 61 GNGSFGVVVFQAKLCDTGLVLAIKKVLQDRRFKNRELQIMRKLEHCNIVKLLYFFYSGEK

seq1: 101 RDEVFLNLVLEIPIETVYKVAR-----FIRLYMYQLFRSLAYIHSLGICHRDI
      |||||
seq2: 121 RDEVFLNLVLEIPIETVYKVARQYAKTKQTIPINFIRLYMYQLFRSLAYIHSLGICHRDI
```

* Legends

- : gap |: identical match *: substitution

Score: Global: Score at the end of the alignment table

Local: Maximum score of the alignment table

Sum: Simply sum of all the scores in the table along the alignment path

Identity: Percent identity (Identical match/(Identical match + Substitution))

* Accessible at

<http://biokdd.informatics.indiana.edu/~juhur/L519/HW5/SeqAlign.html>

* Note on substitution matrixes

1) BLOSUM and PAM Matrixes

They are the standard amino acid substitution scoring matrixes based on the BLOCKS and evolutionary information respectively. They were obtained from the NCBI's [BLAST package](#).

2) Hydrophobicity Matrix

Hydrophobicity scoring matrix constructed from hydrophilicity data. (M. Levitt, J. Mol. Biol. 104, 59 [1976]), derived by George et al. 1990, Mutation Data Matrix and Its Uses, Methods in Enzymology 183, 333. It was obtained from the [web](#).

3) Amino Acid Size Matrix

The matrix was created by me. It's based upon the physical property of amino acids and the source is <http://www.russell.embl-heidelberg.de/aas/aas.html> by Robert B. Russell, Matthew J. Betts & Michael R. Barnes

20 Amino acids were grouped into three categories.

1. Tiny : A, G, C, S
2. Small : P, N, D, T, V
3. Large : E, F, H, I, K, L, Q, R, W, Y, M

The scoring scheme is as following

Self	<->	Self	: 10
Tiny	<->	Tiny	: 5
Tiny	<->	Small	: 2
Tiny	<->	Large	: -5
Small	<->	Small	: 5
Small	<->	Large	: -2
Large	<->	Large	: 5

This scheme is based upon my personal insight and a simple assumption that same group can be acceptably substituted but substitutions between tiny(small) and large groups are generally unfavorable.